Application of the new GeN-Foam multi-physics solver to the European Sodium Fast Reactor and verification against available codes

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Abstract – *A new multi-physics solver for nuclear reactor analysis, named GeN-Foam (Generalized Nuclear Foam), has been developed by the FAST group at the Paul Scherrer Institut. It is based on OpenFOAM® and has been developed for the multi-physics transient analyses of pin-based (e.g., liquid metal Fast Reactors, Light Water Reactors) or homogeneous (e.g., fast spectrum Molten Salt Reactors) nuclear reactors. It includes solution of coarse or fine mesh thermal-hydraulics, thermal-mechanics and neutron diffusion. In particular, thermal-hydraulics solution can combine on the same mesh both a traditional RANS model and a porous medium model, depending on the desired degree of approximation for each region. In case the active reactor core is modeled as a porous medium, a simple sub-solver computes the sub-scale radial temperature profiles in fuel and cladding. The mesh used for neutronics calculations is deformed according to the displacement field predicted by the thermal-mechanics solver, thus allowing for a direct prediction of expansion-related feedback effects in Fast Reactors. To limit computational requirements, GeN-Foam permits the use of three different unstructured meshes for thermal-hydraulics, thermal-mechanics and neutron diffusion. For the same reason, an adaptive time step is employed. The different equations can be solved altogether or selectively included. In this work, GeN-Foam is applied to the analysis of the European Sodium Fast Reactor (ESFR). In particular, a 3-D model of the ESFR core is set up employing a coarse-mesh porous-medium approach for the thermal-hydraulics. The reactor steady-state and different accidental transients are investigated to offer an overview of GeN-Foam use and capabilities, as well as to preliminarily investigate the impact of a relatively accurate thermal-mechanic treatment on the predicted ESFR behavior. A code-to-code benchmark against the TRACE system code is performed to verify the adequacy of the results provided by the new code. GeN-Foam has proved itself a powerful and flexible tool for the ESFR analysis, allowing also to shed some light on phenomena neglected by more simplified treatments. In addition, the code-to-code benchmark showed that GeN-Foam can provide results consistent with state-of-the-art codes, thus giving confidence about the capability of the code to provide adequate predictions of a reactor behavior.*

I. INTRODUCTION

Behavior of nuclear reactors has been simulated for many years using data exchange interfaces between available legacy codes for neutron diffusion and 1-D or sub-channel thermal hydraulics[1,2]. Typically, system non-linearities are not consistently resolved, causing inaccuracies or requiring very short time steps to achieve stable and accurate solutions[3,4]. In addition, most common legacy codes do not support parallel computing, thus limiting the size of problems that can be investigated. The last, perhaps most critical problem associated to the coupling of available codes relates to the difficulties of modification, which is becoming crucial following the need for more accurate simulations[5], the growing interest in particular details of reactor analysis, and the development of innovative reactor systems.

The FAST group at the Paul Scherrer Institut has employed for many years a coupled TRACE-PARCS routine specifically adapted for Fast Reactors (FRs) and validated against relevant experimental data[6]. Following the trends discussed above, a current objective is to develop a new multi-physics platform that can supplement the available FAST code system with a more modern tool. Specific objectives are more implicitly coupled solutions, proper parallelization algorithms, and flexibility in terms of modifications and implementation of new features.

An available numerical library has been used to reduce the development efforts while making use of the latest computational techniques in the field of engineering.

Specifically, the C++ based library OpenFOAM®[7,8] has been selected. This library includes all necessary routines for the discretization and parallel solution of partial differential equations on unstructured meshes using finite-volume methods. Based on OpenFOAM®, a multi-physics solver named GeN-Foam has been developed[9] that semi-implicitly couples fine-mesh or coarse-mesh (porous medium) 3-D thermal-hydraulics, thermal-mechanics, multi-group neutron diffusion, and a simple finite-difference sub-scale fuel model that can be used in coarse-mesh simulations to evaluate the local temperature profile in fuel and cladding.

The objective of this paper is to present briefly the main features of GeN-Foam (which is presented in more details in Ref. 9) and to show its potential use via application to the steady-state and transient analysis of the European Sodium Fast Reactor (ESFR). For code verification purposes, results are compared with those provided by a simple 1-D TRACE[1,10] model.

The paper is organized as follows. Section II presents the GeN-Foam solver. Section III presents the ESFR design and the modeling strategy for both GeN-Foam and TRACE. Section IV presents the results obtained using GeN-Foam and TRACE for the ESFR steady-state and transient behavior. The conclusions of the work are drawn in Section V.

II. THE GeN-Foam MULTI-PHYSICS SOLVER

GeN-Foam is constituted by four main components, namely: a thermal-hydraulics sub-solver; a thermal-mechanics sub-solver; a neutronics sub-solver; and a simple finite-difference 1-D sub-solver that evaluates temperatures in fuel and cladding in case the core is simulated using a coarse mesh. Three unstructured meshes are used for thermal-hydraulics, thermal-mechanics and neutron diffusion, while the sub-scale fuel model is solved in each mesh cell within the fuel zones of the thermal-hydraulics mesh. All sub-solvers have parallel computation capabilities and parallelization is achieved through geometrical domain decomposition. Discretization of the equations is performed according to a finite-volume approach (except for fuel temperatures). Time integration uses a first–order implicit Euler scheme. The next subsections briefly describe the four GeN-Foam main components and the coupling strategy. More details can be found in Ref. 9.

II.A. Thermal-hydraulics

The thermal-hydraulics sub-solver is based on the standard k-ε turbulence model, but extended to coarse-mesh applications through the use of a porous medium approach[11] for user-selected cell zones inside the mesh. An advantage of this treatment is that porous medium equations revert back to standard RANS equations in case of clear fluid, so that the same set of equations can be discretized and solved on the same mesh while treating different zones of the geometry with two different approaches. This alleviates the numerical problems (e.g., of stability or numerical diffusion) that can instead be encountered when trying to couple different solutions from different meshes or from different solvers. Solution of the Navier Stokes equations is achieved through a merged PISO-SIMPLE (named PIMPLE) pressure-based algorithm for compressible flows[7,12,13].

II.B. Sub-scale fuel

In case the reactor core is modelled as a porous structure, it is necessary to employ a sub-scale model for the prediction of temperatures in fuel and cladding. GeN-Foam currently includes a simple sub-solver for a 1-D radial solution of the heat conduction equations in fuel and cladding for each cell containing the fuel. Discretization is performed using a first-order implicit Euler scheme in time and a second-order finite-difference scheme in space[14]. The coupling with the solution for coolant temperature is achieved by calculating the heat flux on the clad surface and transferring it to the coolant in the form of a volumetric heat source via multiplication with the fuel volumetric area. Conversely, the coolant temperature and the heat transfer coefficient between coolant and cladding are used to set up a convective boundary condition for the conduction equation in the cladding. An analogous procedure is used to couple conduction equations for fuel and cladding through definition of a gap conductance and convective (Robin) boundary conditions. The sub-solver includes the possibilities of simulating both solid and annular fuel pins and the number of nodes in fuel and cladding is user selectable.

II.C. Thermal-mechanics

A thermal-mechanic solver named solidDisplacementFoam available in the OpenFOAM® 2.3.0 version has been used as basis for the solution of the thermal-mechanics equations. It is a displacement-based transient segregated finite-volume solver of linear-elastic, small-strain deformation of a solid isotropic body, with optional thermal diffusion and thermal stresses. This thermo-mechanics solver has been slightly modified[9] and is used as a sub-solver of GeN-Foam to evaluate the temperature-induced deformation of the main structures such as vessel, strongback, and diagrid, the latter determining the radial displacement in the core. Axial deformation of the neutronic mesh in the active core is calculated independently, based on the fuel (in case of open gap) or cladding (in case of closed gap) expansion coefficients and temperatures. The same approach is employed to determine the axial displacement of the control rod driveline.

II.D. Neutronics

Solution of the spatial neutron kinetics is based on the traditional multi-group diffusion equations and resolved via Picard iterations. The neutronic sub-solver is capable of time dependent or eigenvalue calculations. The number and structure of both neutron energy and precursors groups is arbitrary. The mesh for neutronics calculations is deformed according to the displacement field calculated by the thermal-mechanic sub-solver.

Different cross-section sets for different perturbed states can be used in each zone of the mesh. Using cell-wise information of the perturbed variable, the sub-solver evaluates the local cross-sections to be used by interpolating between nominal and perturbed cross section sets. For the ESFR, six cross-sections sets are supplied for each cell zone, namely: 1) nominal; 2) increased fuel temperature; 3) axially expanded fuel; 4) radially expanded core; 5) reduced coolant density; 6) expanded cladding. The interpolation is linear in all cases but the fuel temperature, for which a logarithmic interpolation is known to be more suited for FRs[15].

Cross-sections employed are generated using the Monte Carlo code Serpent-2[16] and both a Matlab[17] and an Octave[18] routines heve been implemented to automatically transfer the cross-section into a format suitable for use in OpenFOAM®. For the case of the ESFR, six full core simulations (based on the Serpent input from Ref. 19) have been performed to generate the cross-sections in each zone, adopting homogenization methods recently implemented in Serpent-2[20].

II.E. Coupling strategy

The GeN-Foam multi-physics solver is built upon the main four sub-solver components described in subsections II.A to II.D. The coupling strategy is depicted in Fig. 1. The simulation starts by selecting a time step. Initial time step is given by the user while subsequent time steps are decided based on the most stringent requirement between Courant number condition in the fluid and a maximum allowed power variation which can be selected by the user.

After selection of the time step, GeN-Foam follows the structure of a PIMPLE loop for solving velocity, pressure and energy. Specifically, density (i.e., continuity) equation is first solved, followed by solution of k and ε equations. An optional velocity predictor step is then performed using the pressure field from the previous iteration. A pressure correction equation (see e.g. Ref. 13) is then solved for a specified number of times to correct for non-orthogonality effects. The new pressure field is used to correct the velocity field. This predictor-corrector loop is repeated until the desired degree of convergence is obtained. The energy equation is finally solved and a number of "outer iterations" are performed to resolve the coupling of pressure, velocity and energy.

After achieving the coolant flow and temperature fields, GeN-Foam solves for the temperatures in sub-scale structures, fuel and cladding. It then goes through the thermal-mechanic sub-solver and the neutronic sub-solver, both including their own Picard iterations to solve for the explicit terms. Initial residuals are evaluated before solving each equation and the equation is not solved if these residuals are lower than a user-selected tolerance. If all initial residuals are found to be lower than the desired tolerance, GeN-Foam moves to the next time step.

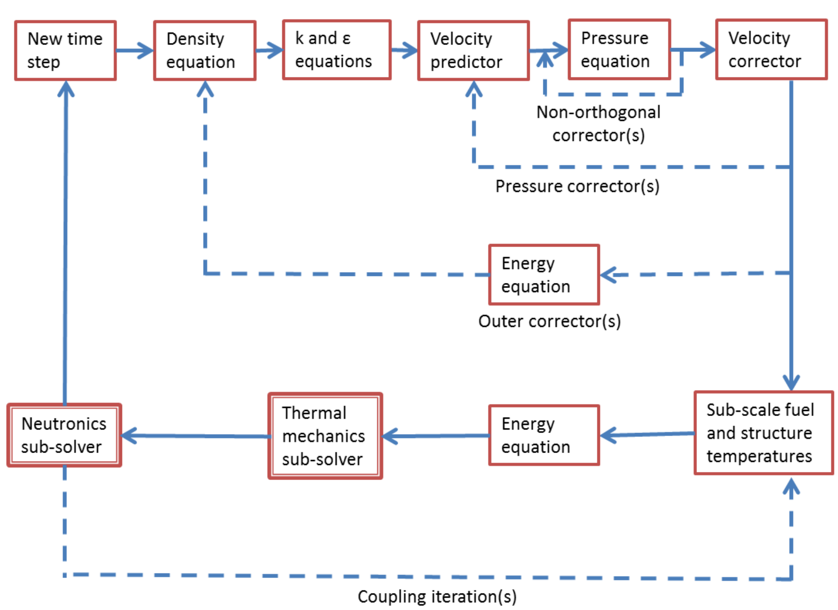

Fig. 1: GeN-Foam coupling strategy[9]

III. THE ESFR DESIGN AND MODELLING STRATEGY

The ESFR (Figs. 2, 3 and 4) is a 3.6 GWth pool-type sodium-cooled FR. It has been designed during the EURATOM ESFR project within the seventh European Framework Program and design details can be found in Refs. 21 and 22. A slightly modified and simplified design has been used in this work that features a 60 degree symmetry in the core configuration (Fig. 4). Tables I and II list the main parameters and physical properties used for simulations in the present paper. It should be noted that constant physical properties are used, but the possibility of temperature dependent properties is included in GeN-Foam.

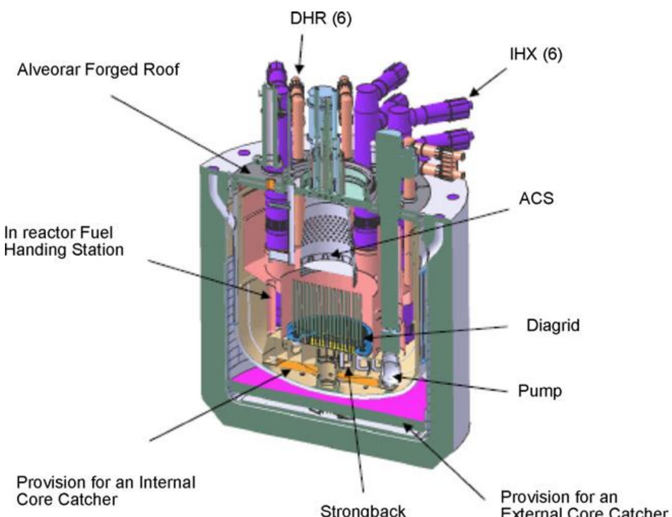

Figure 2: ESFR pool-type concept[21]

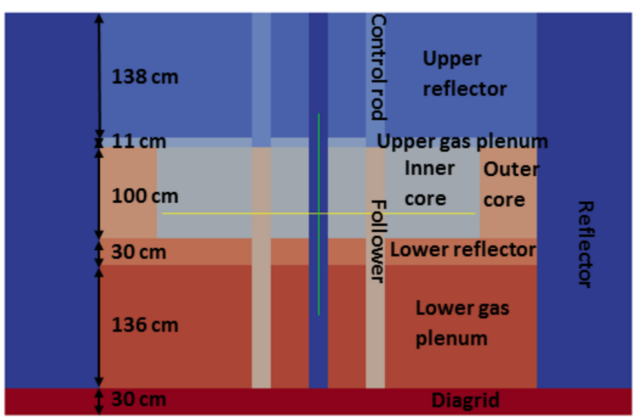

Figure 3: Axial configuration of the reference ESFR core[9]

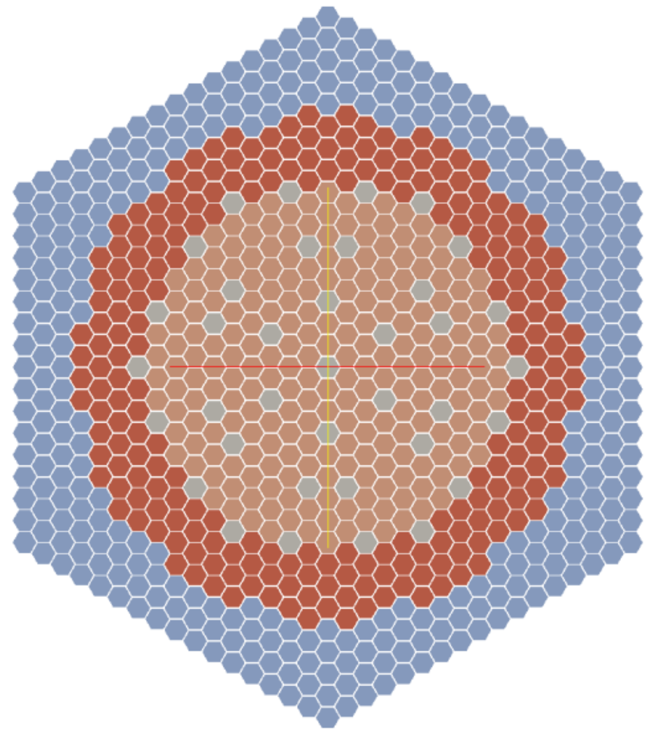

Figure 4: Radial configuration of the reference ESFR core[9]

TABLE I

Nominal conditions and plant parameters used in the ESFR simulations[21,22]

Reactor Power (GWth)	3.6
Number of fuel assemblies	456
Assembly pitch (mm)	210.8
Wrapper flat to flat outer (mm)	206.3
Wrapper flat to flat inner (mm)	197.3
Number of pins per FA	271
Fuel pin outer clad diameter (mm)	10.73
Fuel pin inner clad diameter (mm)	9.73
Fuel pellet outer diameter (mm)	9.43
Fuel pellet inner diameter (mm)	2.4
Cladding material	ODS Steel
Fuel Pellet material	$(U,Pu)O_2$
Core inlet temperature (˚C)	395
Core outlet temperature (˚C)	545
Total mass flow rate through the active core ($kg{\cdot}s^{-1}$)	18940

TABLE II

Material properties used in the ESFR simulations[23,24]

Sodium (743 K)	
Density (kg·m^{-3})	841.06
Specific heat capacity (J·kg^{-1}·K^{-1})	1267.13
Dynamic viscosity (Pa·s)	2.54·10-4
Fuel	
Density (kg·m^{-3})	10480
Specific heat capacity (J·kg^{-1}·K^{-1})	250
Thermal conductivity (W·m^{-1}·K^{-1})	3
Linear expansion coefficient (K^{-1})	1.1·10-5
Cladding	
Density (kg·m^{-3})	7500
Specific heat capacity (J·kg^{-1}·K^{-1})	500
Thermal conductivity (W·m^{-1}·K^{-1})	20
Linear expansion coefficient (K^{-1})	1.8·10-5
Core structures	
Density (kg·m^{-3})	8000
Specific heat capacity (J·kg^{-1}·K^{-1})	600
Thermal conductivity (W·m^{-1}·K^{-1})	20
Linear expansion coefficient (K^{-1})	1.8·10-5
Gap conductance (W·m^{-2}·K^{-1})	3000

III.A. Modelling with GeN-Foam

A 3-D GeN-Foam model has been built using the geometry and mesh shown in Figs. 5 (the same mesh is employed for simplicity for thermal-hydraulics, thermal-mechanics and neutronics), and one-group cross-sections calculated using the Serpent model presented in Ref. 19 (and modified for use in Serpent-2). For simplicity, a radially flat velocity profile has been considered. The entire core is treated as a porous medium. Additional details can be found in Ref. 9.

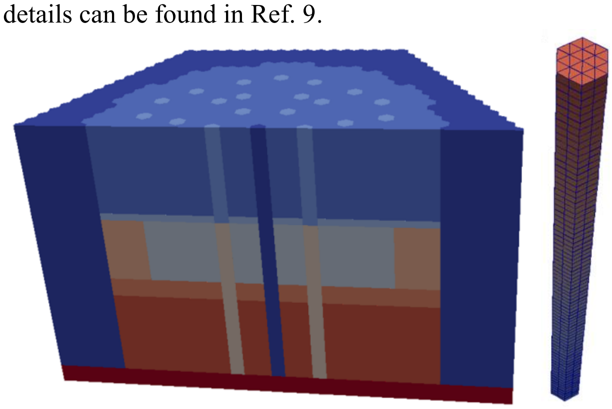

Fig. 5. Geometry (with cell zones) and mesh detail for one assembly for the GeN-Foam model

III.B. Modelling with TRACE

The TRACE model consists of a single fuel rod cell featuring the same average power and coolant velocity as the GeN-Foam model. Neutronics is solved in TRACE using its built-in point-kinetics model, with generation time and Beta effective from adjoint-weighted Serpent-2 calculations, based on the Iterated Fission Probability method[25], and the feedback coefficients obtained through eigenvalue calculations with GeN-Foam (Table 3). Specifically, feedback coefficients have been obtained by comparing results of eigenvalue calculations where the perturbed variable has been uniformly changed in the domain. Axial power density shape has been taken from the Serpent-2 model.

TABLE III

Neutron kinetics parameters used for the TRACE point kinetic model

Generation time	0.507 µs
Beta eff. / precursor decay time	
Group 1	5.8 pcm / 0.0125 s^{-1}
Group 2	61.8 pcm / 0.0283 s^{-1}
Group 3	22.5 pcm / 0.0425 s^{-1}
Group 4	56.4 pcm / 0.133 s^{-1}
Group 5	12.3 pcm / 0.292 s^{-1}
Group 6	54.6 pcm / 0.666 s^{-1}
Group 7	45.1 pcm / 1.63 s^{-1}
Group 8	22.8 pcm / 3.55 s^{-1}
Doppler constant	1094 pcm
Fuel expansion coefficient	-0.331 pcm/K
Radial expansion coefficient	-0.724 pcm/K
Cladding expansion coefficient	+0.147 pcm/K
Coolant expansion coefficient	
Active core	+0.503 pcm/K
Above active core	-0.0825 pcm/K

IV. RESULTS AND DISCUSSION

GeN-Foam and TRACE models have been used to predict the ESFR behavior at steady state and under three main accidental scenarios, namely: UTOP – Unprotected Transient OverPower; ULOF – Unprotected Loss Of Flow; ULOHS - Unprotected Loss Of Heat Sink. The calculations here presented are not to be considered a safety analysis, especially in view of the lack of modelling of the sodium pool. As mentioned, main objectives are to show use and capabilities of GeN-Foam, and to benchmark its results with an available verified code.

Fig. 6 shows the steady-state axial temperatures profiles in 5 different radial locations (coolant bulk, outer surface of the cladding, inner surface of the cladding, outer surface of the fuel, and inner surface of the fuel). Results for GeN-Foam have been obtained from a radial core location featuring a power equal to the average one. The agreement between the two codes is satisfactory. Small discrepancies can be observed and are related to a slightly different axial power shape (obtained from a Monte Carlo calculation and from diffusion theory for TRACE and GeN-Foam, respectively).

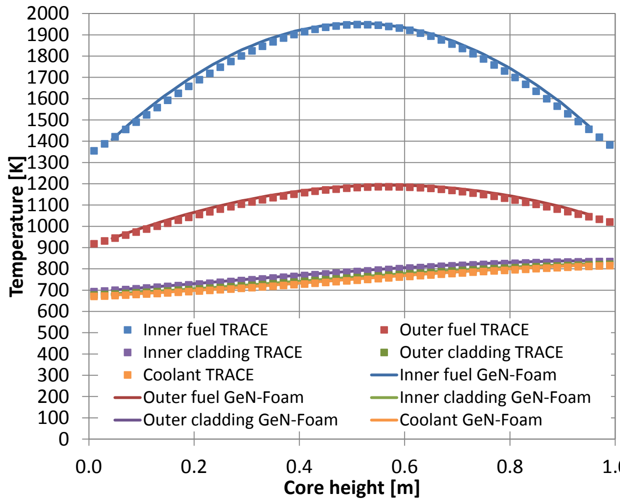

Fig. 6. Comparison of GeN-Foam and TRACE results in terms of average axial temperature profiles

Fig. 7 shows the results obtained using the two codes for a UTOP consisting of a 100 pcm linear reactivity insertion of 2 seconds. Fig. 8 shows the results obtained for a ULOF consisting of an exponential reduction of the flow rate to 20% the nominal value and with half flow rate reached in 10 seconds. Finally, Fig. 9 shows the results obtained for a ULOHS consisting of an exponential increase of the inlet temperature of 50 K, with half of the increment reached in 10 seconds.

A good agreement is observed, although a non-negligible discrepancy appears in case of a ULOF and a UTOP. Observed discrepancies are related to an approximate treatment of reactivity feedbacks in the TRACE model. In fact, one single feedback coefficient is used to describe each reactivity effect, thus neglecting any spatial effect. With such assumption, it would be necessary to employ a different "optimized" set of coefficients for each transient, while in the calculations here performed one single set (Table III), obtained assuming spatially uniform variables, has been considered.

Specifically, an overestimation of the axial expansion coefficient has been found to be main responsible for the different behaviors predicted by TRACE. In fact, the radial power profile in the ESFR is peaked in a peripheral region of the core, where the adjoint flux is relatively low. More important, the power is peaked where there are few control rods (see Fig. 10). Since part of the negative reactivity insertion due to axial expansion derives from the fuel expanding in the region of control rods (that are sitting immediately above the core in the GeN-Foam model), the negative reactivity feedback results lower compared to that predicted by assuming a uniform temperature in the core (which is how the feedback coefficients for TRACE have been derived). To confirm this assumption, the fuel expansion reactivity feedback coefficient has been recalculated by comparing two eigenvalue calculations performed with the core at the initial and final steady state

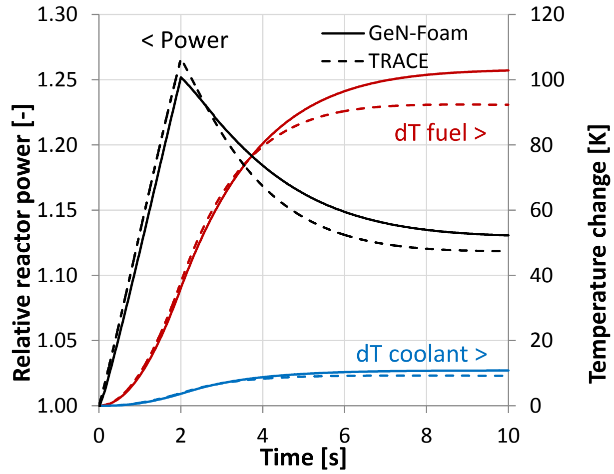

Fig. 7. Comparison of GeN-Foam and TRACE results for a UTOP transient

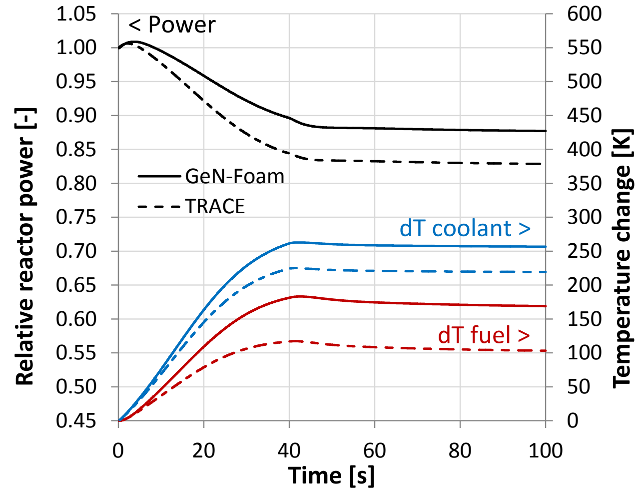

Fig. 8. Comparison of GeN-Foam and TRACE results for a ULOF transient

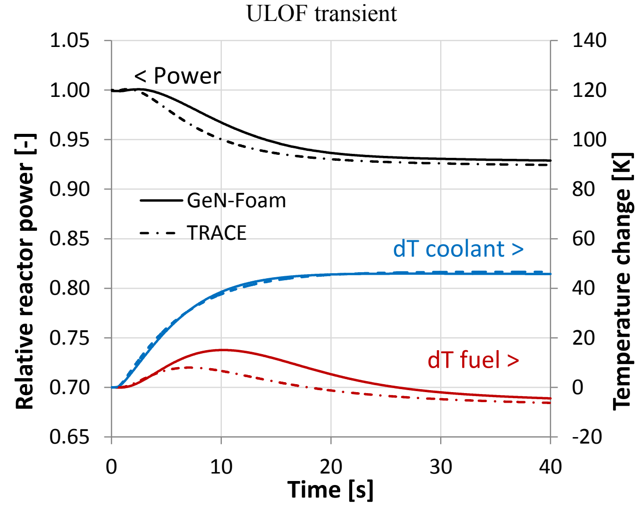

Fig. 9. Comparison of GeN-Foam and TRACE results for a ULOHS transient

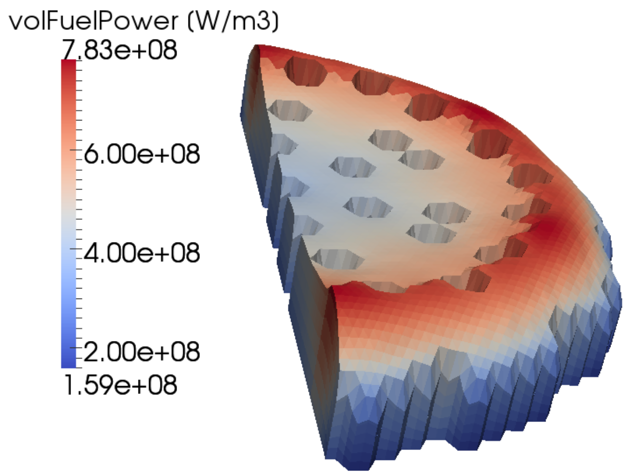

Fig. 10. Fuel volumetric power in the bottom part of the core as predicted by GeN-Foam. Deformation of geometry is magnified by 100 times

of the UTOP transient. All cross section dependencies have been switched off except for fuel expansion. The result is a reactivity difference of 16.2 pcm, which translates into a feedback coefficient of 0.164 pcm/K if considering that the average fuel temperature differs by 98.7 K between the two steady states. The fuel expansion coefficient calculated with a correct temperature profile is thus half the one predicted with a flat temperature profile and used for the calculations in Figs. 7 to 9.

The reason why bigger discrepancies appear in the ULOF and UTOP transients is related to their higher sensitivity to perturbations of the fuel expansion reactivity feedback, as shown in Fig. 11. Results in Fig. 11 have been obtained by performing reactivity balances between initial and final steady states, linearizing the Doppler effect and assuming a symmetric axial power profile. It is also worth noting that a lower axial expansion reactivity feedback is also consistent with the slower power decrement in the ULOHS.

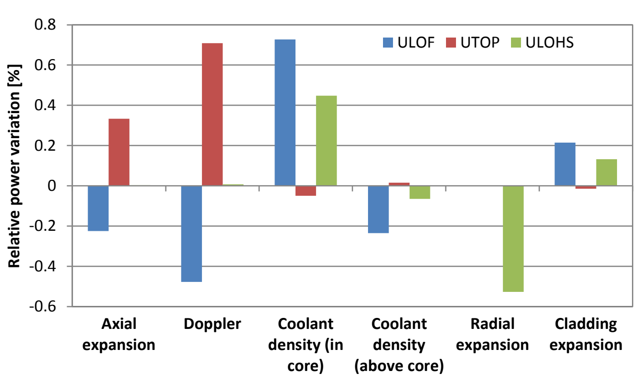

Fig. 11. Sensitivity of the asymptotic reactor power variation to a 1% variation in the feedback coefficients

V. CONCLUSIONS

In this paper, a recently developed multi-physics solver for nuclear reactor analysis, named GeN-Foam, has been briefly presented. It is based on the finite-volume software OpenFOAM® and it includes the solution of coarse or fine mesh thermal-hydraulics, thermal-mechanics and neutron diffusion. The thermal-hydraulics solution combines both a traditional RANS model and a porous medium model on the same mesh, depending on the desired degree of approximation for each zone. A simple sub-solver evaluates the sub-scale radial temperature profiles in fuel and cladding. The mesh used for neutronics calculations is deformed according to the displacement field predicted by the thermal-mechanics solver, thus allowing for a direct and accurate prediction of expansion-related feedback effects in FRs. The different equations can be solved altogether or selectively included and the coupling is semi-implicit with Picard iteration. Parallel computations can be performed via domain decomposition.

After describing the solver, application to the ESFR has been presented. A porous medium description of the ESFR core has been considered and a simple TRACE model has been employed to provide a preliminary benchmark to GeN-Foam results, as well as to point out some specific features of the new solver. Results provided by the two codes show a good agreement both at steady-state and for accidental transient analysis. Discrepancies have been ascribed to simplifications applied in TRACE for simulation of the fuel axial expansion coefficient. The source of such inaccuracy has been found in the methodology employed to evaluate the feedback coefficients, i.e., by comparing the eigenvalues of core states with different but spatially uniform temperatures. In fact, the power shape in the ESFR is peaked in the outer core region, where the adjoint flux is low and where there are few control rods. This causes the axial expansion to be higher where its reactivity effect is lower. The result is an effective axial expansion coefficient that is half the value predicted using spatially uniform temperatures.

To conclude, GeN-Foam has been successfully benchmarked against TRACE result and has proved itself a powerful and flexible tool for FR analysis. A unique feature of this tool relates to the accurate prediction of core deformation and corresponding reactivity feedbacks, which allowed to single out the important effect on reactivity of a radially non-uniform fuel axial expansion. It should be noted that such effect is particularly strong for the beginning-of-life ESFR core here considered, due to its very peaked and relatively unusual power distribution. However the results obtained should be considered as a warning for the use of spatial neutron kinetics models that employ a radially uniform axial core expansion based on the average fuel temperature. It also suggests particular care while choosing the core conditions for the evaluation of feedback coefficients for point kinetic models.

GeN-Foam is currently a complete multi-physics solver that can be used for the analysis of pin-based or homogeneous nuclear reactors. However, several assumptions and simplifications have been made throughout its development[9]. In addition, optimization is still at a very early stage, and implementation of new features is foreseen to be necessary for specific applications. As regards the SFR safety analysis, implementation of a two-phase flow model is a necessary step and will be the subject of future studies for the authors.

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